

RCIM Retuned Reference Archive Closeout Report

Recovered-original RCIM retuned reference archive closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report records the closeout of recovered-original RCIM retuned model exports into the curated `models/paper_reference/rcim_retuned` archive. The closeout accepts only family-direction exports with all 20 target-level ONNX artifacts and all 20 target-level Python pickle artifacts.

Archive Completeness

Direction	Family	Retune Bundle	Eval Bundle	Export Bundle	ONNX	PKL	Exported Errors
forward	SVM	2026-05-09-23-52-16_fw_retune_bundle	2026-05-09-23-52-16_fw_retune_bundle	2026-05-09-23-52-16_fw_retune_bundle	20	20	0
forward	MLP	2026-05-11-16-55-11_fw_retune_bundle	2026-05-11-16-55-11_fw_retune_bundle	2026-05-11-16-55-11_fw_retune_bundle	20	20	0
forward	RF	2026-05-09-10-02-19_fw_retune_bundle	2026-05-09-13-01-21_fw_eval_bundle	2026-05-09-13-01-34_fw_export_bundle	20	20	0
forward	DT	2026-05-13-15-40-16_fw_retune_bundle	2026-05-13-15-40-16_fw_retune_bundle	2026-05-13-15-40-16_fw_retune_bundle	20	20	0
forward	ET	2026-05-11-18-27-16_fw_retune_bundle	2026-05-11-18-27-16_fw_retune_bundle	2026-05-11-18-27-16_fw_retune_bundle	20	20	0
forward	ERT	2026-05-11-18-27-16_fw_retune_bundle	2026-05-11-18-27-16_fw_retune_bundle	2026-05-11-18-27-16_fw_retune_bundle	20	20	0
forward	GBM	2026-05-11-18-27-16_fw_retune_bundle	2026-05-12-18-16-50_fw_eval_bundle	2026-05-12-18-20-41_fw_export_bundle	20	20	0
forward	HGBM	2026-05-11-18-27-16_fw_retune_bundle	2026-05-12-18-16-50_fw_eval_bundle	2026-05-12-18-20-41_fw_export_bundle	20	20	0
forward	XGBM	2026-05-11-16-55-11_fw_retune_bundle	2026-05-11-16-55-11_fw_retune_bundle	2026-05-11-16-55-11_fw_retune_bundle	20	20	0
forward	LGBM	2026-05-12-11-20-54_fw_retune_bundle	2026-05-12-11-20-54_fw_retune_bundle	2026-05-12-11-20-54_fw_retune_bundle	20	20	0
forward	ELM	2026-05-11-18-21-23_fw_retune_bundle	2026-05-11-18-21-23_fw_retune_bundle	2026-05-11-18-21-23_fw_retune_bundle	20	20	0
backward	SVM	2026-05-09-23-42-54_bw_retune_bundle	2026-05-09-23-42-54_bw_retune_bundle	2026-05-09-23-42-54_bw_retune_bundle	20	20	0
backward	MLP	2026-05-11-09-19-07_bw_retune_bundle	2026-05-11-09-19-07_bw_retune_bundle	2026-05-11-09-19-07_bw_retune_bundle	20	20	0
backward	RF	2026-05-09-09-23-24_bw_retune_bundle	2026-05-09-12-31-49_bw_eval_bundle	2026-05-09-12-32-07_bw_export_bundle	20	20	0
backward	DT	2026-05-09-09-21-57_bw_retune_bundle	2026-05-09-09-21-57_bw_retune_bundle	2026-05-09-09-21-57_bw_retune_bundle	20	20	0
backward	ET	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-11-08-07_bw_retune_bundle	20	20	0
backward	ERT	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-11-08-07_bw_retune_bundle	20	20	0
backward	GBM	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-18-22-07_bw_eval_bundle	2026-05-12-18-22-27_bw_export_bundle	20	20	0
backward	HGBM	2026-05-12-11-08-07_bw_retune_bundle	2026-05-12-18-22-07_bw_eval_bundle	2026-05-12-18-22-27_bw_export_bundle	20	20	0
backward	XGBM	2026-05-11-09-19-07_bw_retune_bundle	2026-05-11-09-19-07_bw_retune_bundle	2026-05-11-09-19-07_bw_retune_bundle	20	20	0
backward	LGBM	2026-05-11-16-05-54_bw_retune_bundle	2026-05-11-16-05-54_bw_retune_bundle	2026-05-11-16-05-54_bw_retune_bundle	20	20	0

Direction	Family	Retune Bundle	Eval Bundle	Export Bundle	ONNX	PKL	Exported Errors
backward	ELM	2026-05-11-09-19-07_bw_retune_bundle	2026-05-11-11-38-37_bw_eval_bundle	2026-05-11-11-38-49_bw_export_bundle	20	20	0

Mean Evaluation Metrics

These values come from the accepted Eval stage summaryCrossValidation+_3.8_allFreq.csv files.

Direction	Family	MSE	RMSE	MAE	MAPE
forward	SVM	0.45221	0.2978	0.184519	0.763904
forward	MLP	0.383808	0.29569	0.213553	66411031743924.046875
forward	RF	0.155355	0.171834	0.063698	0.274823
forward	DT	0.155008	0.170755	0.063618	0.274174
forward	ET	0.140874	0.172043	0.067927	0.227937
forward	ERT	0.140874	0.172043	0.067927	0.227937
forward	GBM	0.125222	0.152528	0.077089	0.42525
forward	HGBM	0.125222	0.152528	0.077089	0.42525
forward	XGBM	0.383808	0.29569	0.213553	66411031743924.046875
forward	LGBM	0.127166	0.153622	0.077013	0.348775
forward	ELM	0.430192	0.289775	0.211945	1.171391
backward	SVM	0.469998	0.340568	0.212199	0.685275
backward	MLP	0.31688	0.31866	0.207254	147176760816685.3125
backward	RF	0.067836	0.123949	0.047092	0.245926
backward	DT	0.11403	0.164084	0.060589	0.370306
backward	ET	0.144024	0.18662	0.061084	0.274356
backward	ERT	0.144024	0.18662	0.061084	0.274356
backward	GBM	0.110792	0.16732	0.085725	0.380283
backward	HGBM	0.110792	0.16732	0.085725	0.380283
backward	XGBM	0.31688	0.31866	0.207254	147176760816685.3125
backward	LGBM	0.186807	0.218703	0.145023	0.675751
backward	ELM	0.387312	0.313587	0.215718	0.799429

Forward Retuned Tables

Table 2 - Amplitude MAE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.002912	4.66302e-05	6.04508e-05	7.55757e-05	7.10815e-05	0.000121731	5.12823e-05	0.000696237	0.000698217	0.00024679
MLP	0.014562	0.007778	0.0111	0.007793	0.009778	0.011258	0.011897	0.008305	0.00701	0.015629
RF	0.00351	3.10238e-05	2.31505e-05	3.6618e-05	2.97441e-05	5.5576e-05	1.52476e-05	8.91596e-05	6.69908e-05	4.50935e-05
DT	0.00351	3.10238e-05	2.31505e-05	3.6618e-05	2.97753e-05	5.5576e-05	1.52993e-05	8.91254e-05	6.69908e-05	4.50935e-05
ET	0.003923	3.39405e-05	2.73594e-05	3.96039e-05	2.81046e-05	6.39258e-05	1.5678e-05	6.23185e-05	6.8023e-05	4.45344e-05
ERT	0.003923	3.39405e-05	2.73594e-05	3.96039e-05	2.81046e-05	6.39258e-05	1.5678e-05	6.23185e-05	6.8023e-05	4.45344e-05
GBM	0.002618	2.63076e-05	1.86476e-05	2.34801e-05	2.56511e-05	2.36904e-05	1.22957e-05	0.000107198	0.000119594	3.49172e-05
HGBM	0.002618	2.63076e-05	1.86476e-05	2.34801e-05	2.56511e-05	2.36904e-05	1.22957e-05	0.000107198	0.000119594	3.49172e-05
XGBM	0.014562	0.007778	0.0111	0.007793	0.009778	0.011258	0.011897	0.008305	0.00701	0.015629
LGBM	0.002575	2.62664e-05	1.87327e-05	2.33772e-05	2.5798e-05	2.51835e-05	1.20466e-05	0.000107466	0.000118095	3.52105e-05
ELM	0.008394	2.88142e-05	7.87396e-05	0.000112282	4.01642e-05	0.000296962	2.83329e-05	0.000856622	0.001067	0.000369496

Table 3 - Amplitude RMSE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.004123	6.08873e-05	7.62318e-05	9.55895e-05	8.97431e-05	0.000165775	6.03122e-05	0.001528	0.002377	0.000612235
MLP	0.023068	0.013469	0.02176	0.015826	0.018224	0.021632	0.020561	0.015097	0.014136	0.025682
RF	0.004879	4.30608e-05	3.33725e-05	5.05612e-05	4.50903e-05	7.97431e-05	2.37782e-05	0.000298279	0.000196766	7.18479e-05
DT	0.004879	4.30523e-05	3.33725e-05	5.05612e-05	4.50974e-05	7.97431e-05	2.38369e-05	0.000298278	0.000196766	7.18479e-05
ET	0.005512	4.98162e-05	4.46899e-05	5.37782e-05	4.20959e-05	8.92871e-05	2.55713e-05	0.000190094	0.000186599	6.97557e-05
ERT	0.005512	4.98162e-05	4.46899e-05	5.37782e-05	4.20959e-05	8.92871e-05	2.55713e-05	0.000190094	0.000186599	6.97557e-05
GBM	0.003898	3.66942e-05	2.64492e-05	3.2727e-05	3.5704e-05	3.51107e-05	1.92161e-05	0.000264253	0.000275736	7.01425e-05
HGBM	0.003898	3.66942e-05	2.64492e-05	3.2727e-05	3.5704e-05	3.51107e-05	1.92161e-05	0.000264253	0.000275736	7.01425e-05
XGBM	0.023068	0.013469	0.02176	0.015826	0.018224	0.021632	0.020561	0.015097	0.014136	0.025682
LGBM	0.00379	3.66354e-05	2.67199e-05	3.21329e-05	3.57956e-05	3.707e-05	1.89939e-05	0.000267423	0.000266197	7.02233e-05
ELM	0.010659	3.82909e-05	9.6453e-05	0.000153107	6.26075e-05	0.0003916	3.88056e-05	0.00157	0.002357	0.000636913

Table 4 - Phase MAE

Model	1	3	39	40	78	81	156	162	240
SVM	0.002861	0.039126	0.038193	0.095298	0.19446	0.167816	1.545304	0.854949	0.747385
MLP	0.020549	0.068079	0.081528	0.07124	0.145159	0.141555	1.653017	0.90848	0.781412
RF	0.002236	0.02734	0.032261	0.045518	0.075769	0.06362	0.490024	0.24597	0.287325
DT	0.002239	0.02734	0.032261	0.045456	0.07449	0.06362	0.490408	0.245312	0.287325
ET	0.002979	0.032667	0.032921	0.046145	0.105701	0.077808	0.474651	0.284212	0.297142
ERT	0.002979	0.032667	0.032921	0.046145	0.105701	0.077808	0.474651	0.284212	0.297142
GBM	0.001955	0.025438	0.020247	0.037234	0.071251	0.048207	0.605127	0.337684	0.391634
HGBM	0.001955	0.025438	0.020247	0.037234	0.071251	0.048207	0.605127	0.337684	0.391634
XGBM	0.020549	0.068079	0.081528	0.07124	0.145159	0.141555	1.653017	0.90848	0.781412
LGBM	0.001928	0.025764	0.020305	0.03646	0.073436	0.045802	0.603746	0.346318	0.383523
ELM	0.002771	0.080557	0.089368	0.067536	0.19027	0.174713	1.703741	1.116297	0.80237

Table 5 - Phase RMSE

Model	1	3	39	40	78	81	156	162	240
SVM	0.00436	0.054297	0.060574	0.132484	0.362591	0.241217	2.02453	1.654005	1.412756
MLP	0.032351	0.086326	0.104059	0.103366	0.212569	0.179317	1.994397	1.406654	1.202815
RF	0.003114	0.039113	0.060054	0.070686	0.172907	0.095694	1.225761	0.891621	0.872011
DT	0.003115	0.039113	0.060054	0.070673	0.151314	0.095694	1.225799	0.891606	0.872011
ET	0.004395	0.049717	0.05807	0.071786	0.294988	0.130349	1.092873	0.927118	0.805299
ERT	0.004395	0.049717	0.05807	0.071786	0.294988	0.130349	1.092873	0.927118	0.805299
GBM	0.002681	0.036307	0.032742	0.057628	0.139074	0.070305	1.03431	0.784113	0.888707
HGBM	0.002681	0.036307	0.032742	0.057628	0.139074	0.070305	1.03431	0.784113	0.888707
XGBM	0.032351	0.086326	0.104059	0.103366	0.212569	0.179317	1.994397	1.406654	1.202815
LGBM	0.002673	0.03642	0.032488	0.056709	0.145076	0.067668	1.049463	0.794237	0.883119
ELM	0.004207	0.099629	0.112824	0.098927	0.285837	0.233381	2.031703	1.636152	1.276826

Backward Retuned Tables

Table 2 - Amplitude MAE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.003417	7.05621e-05	6.44134e-05	0.000108184	3.99598e-05	0.000140876	3.55141e-05	0.000759711	0.000512494	0.000342361
MLP	0.016062	0.008565	0.02068	0.009938	0.009976	0.007135	0.010629	0.008277	0.011534	0.007464
RF	0.002937	2.37386e-05	2.07275e-05	1.72732e-05	2.76117e-05	5.1867e-05	1.00356e-05	0.000117828	9.1534e-05	6.23487e-05
DT	0.003504	2.69959e-05	3.42058e-05	2.59035e-05	3.23554e-05	0.000105717	1.36692e-05	0.000191057	0.000165638	0.000125839
ET	0.003322	2.65069e-05	2.63791e-05	2.03945e-05	3.03482e-05	7.00267e-05	9.3832e-06	0.000111551	9.2988e-05	0.000129119
ERT	0.003322	2.65069e-05	2.63791e-05	2.03945e-05	3.03482e-05	7.00267e-05	9.3832e-06	0.000111551	9.2988e-05	0.000129119
GBM	0.003336	3.32158e-05	2.92694e-05	2.23226e-05	2.9316e-05	8.58519e-05	1.20384e-05	0.000355944	0.000276624	0.000115003
HGBM	0.003336	3.32158e-05	2.92694e-05	2.23226e-05	2.9316e-05	8.58519e-05	1.20384e-05	0.000355944	0.000276624	0.000115003
XGBM	0.016062	0.008565	0.02068	0.009938	0.009976	0.007135	0.010629	0.008277	0.011534	0.007464
LGBM	0.006586	3.53015e-05	4.62261e-05	4.03759e-05	3.28733e-05	0.000230488	1.56541e-05	0.000577349	0.000432744	0.000209221
ELM	0.006105	4.0656e-05	6.19064e-05	6.94347e-05	3.74189e-05	0.000314255	2.18478e-05	0.001246	0.000861271	0.000374497

Table 3 - Amplitude RMSE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.004321	8.92984e-05	7.59251e-05	0.000122372	5.24986e-05	0.00018634	4.13104e-05	0.002147	0.001733	0.000768566
MLP	0.023489	0.014012	0.034222	0.016157	0.019279	0.01215	0.018511	0.011909	0.020039	0.01354
RF	0.003822	4.01928e-05	3.95792e-05	2.41367e-05	4.08393e-05	7.35254e-05	1.45444e-05	0.000351421	0.000350566	0.000143677
DT	0.004755	3.89166e-05	4.63022e-05	3.5792e-05	4.65966e-05	0.000141259	2.1564e-05	0.000458745	0.000633652	0.000247847
ET	0.004572	3.78036e-05	4.05121e-05	3.04868e-05	4.2643e-05	9.56236e-05	1.38535e-05	0.000333265	0.000508611	0.000340328
ERT	0.004572	3.78036e-05	4.05121e-05	3.04868e-05	4.2643e-05	9.56236e-05	1.38535e-05	0.000333265	0.000508611	0.000340328
GBM	0.004305	5.05026e-05	4.01874e-05	3.17414e-05	4.19043e-05	0.000115062	1.73374e-05	0.00091268	0.000819282	0.000242704
HGBM	0.004305	5.05026e-05	4.01874e-05	3.17414e-05	4.19043e-05	0.000115062	1.73374e-05	0.00091268	0.000819282	0.000242704
XGBM	0.023489	0.014012	0.034222	0.016157	0.019279	0.01215	0.018511	0.011909	0.020039	0.01354
LGBM	0.008224	5.37816e-05	5.75288e-05	5.02789e-05	4.44481e-05	0.000274288	2.08906e-05	0.00103	0.000972133	0.000415477
ELM	0.007959	6.49451e-05	8.87046e-05	8.60459e-05	4.79292e-05	0.000405392	2.81136e-05	0.002354	0.001667	0.000692278

Table 4 - Phase MAE

Model	1	3	39	40	78	81	156	162	240
SVM	0.003277	0.030147	1.533948	0.239405	0.16531	0.165708	0.519181	0.647235	0.934282
MLP	0.012577	0.047895	0.883528	0.223007	0.117947	0.164507	0.550536	0.633432	0.74779
RF	0.004475	0.020274	0.259613	0.088339	0.055395	0.100364	0.103195	0.104946	0.201875
DT	0.002297	0.030949	0.264166	0.109481	0.104082	0.131394	0.154214	0.138677	0.272296
ET	0.002216	0.024758	0.251627	0.12914	0.061557	0.117384	0.137711	0.158073	0.335381
ERT	0.002216	0.024758	0.251627	0.12914	0.061557	0.117384	0.137711	0.158073	0.335381
GBM	0.004355	0.031669	0.48239	0.11461	0.089437	0.116216	0.251079	0.198438	0.422011
HGBM	0.004355	0.031669	0.48239	0.11461	0.089437	0.116216	0.251079	0.198438	0.422011
XGBM	0.012577	0.047895	0.883528	0.223007	0.117947	0.164507	0.550536	0.633432	0.74779
LGBM	0.003988	0.075939	1.007733	0.159638	0.119033	0.16781	0.347521	0.403988	0.606608
ELM	0.005351	0.095946	1.401709	0.21804	0.14142	0.191291	0.566196	0.85855	0.826733

Table 5 - Phase RMSE

Model	1	3	39	40	78	81	156	162	240
SVM	0.004976	0.050379	1.904003	0.390645	0.302424	0.208329	0.96709	1.234879	1.739106
MLP	0.019934	0.061518	1.488289	0.387442	0.157908	0.216736	0.922763	1.028735	1.240815
RF	0.017312	0.031548	0.925908	0.133734	0.089557	0.134517	0.321643	0.324612	0.495247
DT	0.00395	0.046179	1.158681	0.159423	0.153046	0.187845	0.411266	0.488621	0.666233
ET	0.00377	0.037153	1.108501	0.245136	0.110455	0.16568	0.50731	0.556874	0.991502
ERT	0.00377	0.037153	1.108501	0.245136	0.110455	0.16568	0.50731	0.556874	0.991502
GBM	0.008257	0.042245	1.003621	0.175671	0.164139	0.159327	0.552074	0.448768	0.785715
HGBM	0.008257	0.042245	1.003621	0.175671	0.164139	0.159327	0.552074	0.448768	0.785715
XGBM	0.019934	0.061518	1.488289	0.387442	0.157908	0.216736	0.922763	1.028735	1.240815
LGBM	0.006405	0.089842	1.316174	0.239411	0.186768	0.208583	0.68118	0.644011	0.990548
ELM	0.011728	0.12662	1.852693	0.376851	0.186686	0.249883	0.946219	1.157153	1.350511

Source Recovery Notes

- RF uses separate backward Eval and Export recovery bundles after the parser fix.
- ELM uses later recovery export bundles where the custom ONNX exporter is active.
- GBM and HGBM use later recovery export bundles that removed earlier ONNX export errors.
- LGBM uses the quieter retune factory surface so native LightGBM chatter does not bury failure evidence.